SCHEDULERLENS: CAUSAL INTERPRETABILITY OF MASKED DIFFUSION LANGUAGE MODELS AND UNMASKING SCHEDULERS

Anonymous authors
Paper under double-blind review

ABSTRACT

[Devi: placeholder for now: Don't review] Masked Diffusion Language Models (MDLMs) generate text by iteratively unmasking multiple token positions in parallel, making generation quality jointly dependent on the underlying model and the scheduler that selects which tokens to reveal. Existing work typically evaluates models and schedulers independently, leaving unexplained why the same scheduler can perform well for one MDLM yet poorly for another. We introduce SCHEDULERLENS, a causal framework for analyzing model–scheduler interactions through a generation-error decomposition that separates prediction error from parallelism error. The framework identifies distinct pathways through which model and scheduler changes affect decoding behavior and introduces observable trajectory-level diagnostics for studying these effects without access to oracle data distributions. Across 315 runs spanning 3 MDLMs, 7 schedulers, 5 benchmarks, and 3 decoding budgets, we find that scheduler effectiveness depends not only on the scheduler itself but on how its implicit assumptions about token dependencies align with a model's learned representations and the structure of the task.

1 INTRODUCTION

Masked Diffusion Language Models (MDLMs) extend diffusion models from continuous data to discrete text token sequences (Austin et al., 2021; Sahoo et al., 2024; Nie et al., 2026). Instead of traditional autoregressive left-to-right decoding, MLDMs begin with a fully masked sequence and iteratively denoise it, unmasking multiple tokens at arbitrary, non-contiguous positions across the sequence in parallel at each step. Generation needs (1) a model that predicts a distribution for each masked position and (2) a scheduler that chooses how many tokens to unmask and which ones. The challenge is that tokens revealed in the same step are typically sampled independently from their predicted distributions, even though the underlying text contains strong dependencies among them. As a result, unmasking mutually dependent tokens simultaneously can degrade sample quality, even with a strong model. Avoiding this would mean knowing where the dependencies lie, which is a property of the true data distribution the scheduler cannot directly access, so it depends on inference time proxies. Some methods use model-dependent uncertainty at inference time (e.g., confidence, margin and entropy); others enforce a fixed prior through human defined rules or policy functions. These are alternative estimators of the same latent quantity, whether a set of tokens is safe to reveal/unmask together given what is already committed. This leads us to the question:

> *Why does the same scheduler perform well for one masked diffusion language model but poorly for another on a particular task, and what prediction and selection mechanisms drive this interaction?*

To address the question, we must focus on how models and schedulers interact as a pair, rather than trying to rank models or schedulers in isolation. A scheduler that works well with one model but poorly with another is not simply evidence of a good or bad scheduler. Scheduler-focused work typically compares decoding policies while treating models independently (Arriola et al., 2025; Ben-Hamu et al., 2026; Luxembourg et al., 2026; Mo et al., 2025), while model-focused work compares checkpoints or training methods while treating schedulers independently (Nie et al., 2026; Ye et al.,

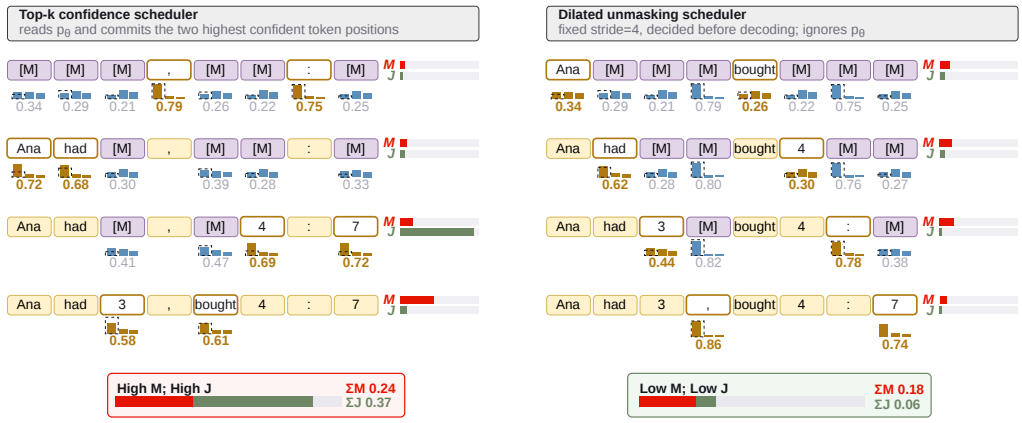


Figure 1: Step-by-step generation using two different schedulers but same model (p_θ). Each step commits two positions. At masked positions, bars show p_θ and dashed outlines the true marginals q. (Left) Top-k confidence scheduler ranks prediction marginals and unmasks top-k tokens; at step 3 it reveals "4" and "7", whose marginals are flat but whose joint is not, incurring huge parallelism error $\mathcal{J}_t|_{t=3}$. By step 4, q at "3" is high but the model is less confident, causing prediction error $\mathcal{M}_t|_{t=4}$. (Right) Dilated scheduling fixes the stride in advance and reaches "7" only after "3" and "4" are unmasked, so both errors are insignificant.

2025; Zhu et al., 2025). Each approach is defensible on its own, but neither can reveal the effects that only emerge when both the model and the scheduler vary together. We investigate the interaction and examine how a specific model and a specific scheduler jointly affect the downstream performance.

Let $q(x)$ denote the data distribution over complete token sequences $x = (x_1, \ldots, x_L)$ with length L. An MDLM generates a sequence through a sequence of reverse steps, starting from a fully masked sequence. At step t, let X_t denote the partially unmasked sequence, including the positions and values of all tokens revealed so far. For each position i that remains masked, the model predicts a marginal distribution $p_\theta(x_i \mid X_t)$. A scheduler ϕ then selects a set of masked positions $\mathcal{S}_t$ to reveal in parallel, with each selected token sampled independently from its predicted marginal. Let $p_{\theta,\phi}(x)$ denote the resulting distribution over generated sequences, and so the generation error becomes $D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\phi}\right]$ where $D_{\mathrm{KL}}\left[\cdot \,\|\, \cdot\right]$ denotes the Kullback-Leibler (KL) divergence.

For a deterministic scheduler that never remasks a revealed token, Ben-Hamu et al. (2026) show that the generation error can be decomposed into prediction and parallelism errors:

$$D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\phi}\right] = \mathbb{E}_{x\sim q} \sum_t \left(\underbrace{\sum_{i\in\mathcal{S}_t} D_{\mathrm{KL}}\left[q(x_i \mid X_t) \,\|\, p_\theta(x_i \mid X_t)\right]}_{\mathcal{M}_t\text{: Prediction error}} + \underbrace{D_{\mathrm{KL}}\left[q(x_{\mathcal{S}_t} \mid X_t) \,\|\, \prod_{i\in\mathcal{S}_t} q(x_i \mid X_t)\right]}_{\mathcal{J}_t\text{: Parallelism error}} \right). \tag{1}$$

The prediction error $\mathcal{M}_t$ measures the discrepancy between the model's predicted marginal $p_\theta(x_i \mid X_t)$ at each selected position and the corresponding true conditional marginal $q(x_i \mid X_t)$ for all tokens x_i which have been chosen to be unmasked. The parallelism error $\mathcal{J}_t$ measures the conditional dependence $q(x_{\mathcal{S}_t} \mid X_t)$ among the tokens $\mathcal{S}_t$ selected for simultaneous unmasking that is ignored when they are sampled independently. In particular, $\mathcal{J}_t = 0$ if and only if the tokens in $\mathcal{S}_t$ are conditionally independent given X_t. Thus, even with exact marginal predictions, generation error can persist when conditionally dependent positions are unmasked in parallel. Fig. 1 shows two full examples of variation in $\mathcal{M}_t$ and $\mathcal{J}_t$ during step-by-step generation using two different schedulers but same model (p_θ).

The generation-error decomposition explains why quality can vary across decoding trajectories, but it does not explain why the same scheduler may perform well for one MDLM and poorly for another. We argue that model-scheduler interactions arise through two broad mechanisms: (1) changes in token prediction quality reflected in the prediction-error term $\mathcal{M}_t$ and (2) changes in the sets of

positions selected for parallel unmasking, which in turn affect the parallelism-error term $\mathcal{J}_t$. Because both the model p_θ and scheduler ϕ can influence the selection sets $\mathcal{S}_t$ and the conditioning histories X_t encountered during decoding, comparing either component in isolation can obscure the source of performance differences. We therefore develop a causal framework that decomposes differences in generation error $D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\phi}\right]$ across model–scheduler pairs into effects attributable to the predictor p_θ and effects attributable to the selection process induced by ϕ. This enables a principled analysis of how prediction quality and parallel-unmasking decisions jointly determine downstream generation performance.

Computing $\mathcal{M}_t$ and $\mathcal{J}_t$ requires oracle data conditionals that cannot be measured directly on natural text. We therefore introduce observable diagnostic proxies $\hat{\mathcal{M}}_t$ and $\hat{\mathcal{J}}_t$. These proxies are empirically computed using properties of the decoding trajectories (like model confidence and entropy). These diagnostics serve as interpretable indicators of prediction uncertainty and parallel decoding behavior, allowing us to connect the theoretical decomposition to measurable decoding traces.

Across 315 runs spanning 7 schedulers, 3 MDLMs, 5 benchmarks, and 3 decoding budgets, we investigate how model, scheduler, and dataset characteristics influence downstream performance through the prediction and parallelism errors, $\mathcal{M}_t$ and $\mathcal{J}_t$. Our results show that a scheduler should not be viewed merely as a mechanism for controlling parallelism error. Rather, each scheduler embodies an implicit hypothesis about which token positions can be safely revealed together. Strong performance emerges only when this hypothesis aligns with both the model's learned representations and the dependency structure required by the task, motivating a causal analysis of model–scheduler interactions rather than evaluating either component in isolation.

Overall, we make the following main contributions in this paper: **[Devi: Placeholder: please dont review th contributions]**

1. **A causal framework for MDLM scheduler interpretability.** We introduce SCHEDULER-LENS, a causal framework that explains why model–scheduler interactions arise in masked diffusion language models. The framework decomposes differences in generation quality into distinct pathways attributable to prediction behavior and scheduler-induced token selection.
2. **Theoretical and empirical analysis of prediction and parallelism errors.** Building on the generation-error decomposition of MDLMs, we characterize how scheduler decisions influence prediction error ($\mathcal{M}_t$) and parallelism error ($\mathcal{J}_t$), and show how differences in selection sets mediate model–scheduler interactions during decoding.
3. **Observable diagnostics and large-scale evaluation.** We introduce practical trajectory-level diagnostics based on model confidence and entropy that serve as observable proxies for otherwise inaccessible error quantities. Using 315 runs across 3 MDLMs, 7 schedulers, 5 benchmarks, and 3 decoding budgets, we demonstrate that scheduler effectiveness depends on alignment between scheduler assumptions, model representations, and task-specific dependency structure.

2 PRELIMINARIES ON MDLMS

Masked Diffusion Language Models (MDLMs) generate text by iteratively revealing tokens from an initially fully masked sequence. At decoding step t, let X_t denote the current partially revealed sequence and let z_t be the set of positions that remain masked. For each masked position $i \in z_t$, the model predicts a marginal distribution $p_\theta(x_i \mid X_t)$. A scheduler ϕ then selects a subset of masked positions $\mathcal{S}_t \subseteq z_t$ to reveal. Tokens at the selected positions are sampled independently from their predicted marginals and unmasked, producing the next state X_{t+1}.

Different schedulers use different criteria for choosing $\mathcal{S}_t$. Some rely on model-dependent signals such as confidence (Fig. 1(Left)), margin, or entropy, whereas others follow fixed reveal patterns that do not depend on model predictions (Fig. 1(Right)). Decoding proceeds until all positions have been revealed.

A key property of MDLMs is that the model only predicts token marginals, while the scheduler determines the order and grouping in which positions are revealed. Consequently, generation quality depends on both the predictor and the scheduler, making their interaction a natural object of study.

In our experiments we study three representative MDLMs: LLaDA (Nie et al., 2026), LLaDA-1.5 (Zhu et al., 2025), and Dream 7B (Ye et al., 2025). LLaDA (Nie et al., 2026) is trained from scratch for masked token prediction. LLaDA-1.5 (Zhu et al., 2025) further improves LLaDA using preference-based fine-tuning. Dream 7B (Ye et al., 2025) initializes from an autoregressive language model and uses an adaptive noise schedule during training. These models differ substantially in how they are initialized and trained, providing a useful testbed for understanding why the same scheduling strategy can behave differently across MDLMs.

3 SCHEDULERLENS: A CAUSAL FRAMEWORK FOR MDLM SCHEDULERS

A scheduler determines which token positions are revealed at each decoding step and, consequently, which model predictions become part of the future context. As a result, differences between two model–scheduler pairs can arise from three sources: differences in the underlying token predictor, differences in how model predictions influence reveal decisions, or differences in the scheduler decoding policy itself. We formalize these effects through a causal decomposition of generation error. At decoding step $t \in \{1, \dots, T\}$, let $z_t \subseteq [L]$ denote the set of masked positions, with $z_1 = [L]$ and $z_{T+1} = \emptyset$. The positions revealed at step t are $\mathcal{S}_t = z_t \setminus z_{t+1}$, and X_t denotes the partially revealed sequence before the reveal decision. Let q denote the data distribution and $p_{\theta,\phi}$ the distribution induced by model p_θ together with scheduler ϕ.

Definition 1 (Scheduler)**.** A scheduler ϕ maps the current decoding state X_t to a set of positions $\mathcal{S}_t \subseteq z_t$ to reveal.

A scheduler is *deterministic* if, for a given decoding state X_t, it always returns the same selection set $\mathcal{S}_t = \phi(X_t, p_\theta(\cdot|X_t))$. Otherwise, it is *stochastic* and defines a distribution over possible selection sets.

A scheduler is *open* if the selected positions are independent of model predictions and committed token values in X_t. Otherwise it is *closed*. Open schedulers therefore produce the same reveal sequence across models given the same initial state, whereas closed schedulers may adapt their decisions to model outputs.

Generation Error Decomposition. For a deterministic scheduler that independently samples revealed tokens from p_θ, generation error admits the decomposition (Ben-Hamu et al., 2026)

$$D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\phi}\right] = \mathbb{E}_q\left[\sum_{t=1}^{T}\left(\mathcal{M}_t(\mathcal{S}_t;\theta) + \mathcal{J}_t(\mathcal{S}_t)\right)\right]. \tag{2}$$

The prediction error

$$\mathcal{M}_t(S_t;\theta) = \sum_{i\in S_t} D_{\mathrm{KL}}\left[q(x_i \mid X_t) \,\|\, p_\theta(x_i \mid X_t)\right] \tag{3}$$

measures inaccuracies in model marginals. The parallelism error

$$\mathcal{J}_t(S_t) = D_{\mathrm{KL}}\left[q(x_{S_t} \mid X_t) \,\|\, \prod_{i\in S_t} q(x_i \mid X_t)\right] \tag{4}$$

measures conditional dependence ignored when multiple positions are revealed independently at the same step. Importantly, low prediction error does not imply low parallelism error. Moreover, for a fixed history X_t and selection set S, $\mathcal{J}_t(S)$ depends only on the data distribution and not on model probabilities.

The decomposition suggests that model–scheduler interactions arise through two mechanisms. A model may change the prediction error directly through better token marginals, and it may alter the selected positions under a closed scheduler, thereby changing the parallelism error. SCHEDULERLENS makes these pathways explicit through a causal channel decomposition.

Channel Decomposition. Let

$$\pi_{\theta,\phi}(X) = \phi(X, p_\theta(\cdot|X))$$

denote the reveal policy induced by model θ and scheduler ϕ. Note that we have dropped the notation t for sake of clarity. Define

$$D(\theta, \phi) = D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\phi}\right].$$

For two model–scheduler pairs (θ, ϕ) and (θ', ϕ'),

$$\begin{aligned} D(\theta', \phi') - D(\theta, \phi) = & \underbrace{D(\theta'; \pi_{\theta,\phi}) - D(\theta; \pi_{\theta,\phi})}_{\text{prediction channel}} \\ & + \underbrace{D(\theta'; \pi_{\theta',\phi}) - D(\theta'; \pi_{\theta,\phi})}_{\text{control channel}} \\ & + \underbrace{D(\theta'; \pi_{\theta',\phi'}) - D(\theta'; \pi_{\theta',\phi})}_{\text{structural channel}}. \end{aligned} \tag{5}$$

The structural channel vanishes when $\phi' = \phi$; the control channel vanishes when ϕ is open.

The prediction channel measures the effect of replacing the predictor while holding the reveal policy fixed. The control channel captures changes in reveal decisions caused by altered model scores under a closed scheduler. The structural channel captures changes due solely to modifying the scheduler unmasking policy. The control channel vanishes for open schedulers, while the structural channel vanishes when the scheduler is unchanged.

Selection-set mechanisms. The parallelism error depends only on the selected set. Consequently, both the control and structural channels influence generation quality through changes to the selected positions. We show that any change in a selection set can be decomposed into expansions, contractions, and replacements, each inducing predictable changes in parallelism error. Detailed results and scheduler-specific analyses are provided in Appendix X.

Selection-set mechanisms of the control and structural channels: For a fixed history X, $\mathcal{J}_X(S)$ is determined solely by the selection set S. Thus, a model modification via the control channel, or a scheduler-rule modification via the structural channel, may alter the local parallelism error by changing this set. The atomic types of set changes are one-position expansion and replacement:

$$\mathcal{J}_X(S \cup \{i\}) - \mathcal{J}_X(S) = I_q(x_i; x_S \mid X) \geq 0, \qquad i \notin S, \tag{6}$$

$$\mathcal{J}_X(C \cup \{b\}) - \mathcal{J}_X(C \cup \{a\}) = I_q(x_b; x_C \mid X) - I_q(x_a; x_C \mid X), \qquad a, b \notin C. \tag{7}$$

Enlarging S cannot reduce $\mathcal{J}_X$, and it strictly increases it when the newly added position depends on the existing set. Contraction works in the opposite direction: deleting i changes the error by $-I_q(x_i; x_{S\setminus\{i\}} \mid X) \leq 0$. Replacements may shift $\mathcal{J}_X$ upward or downward. Any modification with a fixed history that takes S to S' can be expressed as a sequence of these atomic operations. We further understand how the various schedulers studied in this work directly or indirectly influence the selection set through the structural channel.

Observable proxies for causal interpretability: Taken jointly, decomposing the process into channels and assigning generation error to separate channels arising from the interaction between the model and the scheduler clarifies what our causal experiments must capture: how changes to a predictor or to a scheduling rule alter marginal prediction error, the resulting selection sets, and the ways these quantities interrelate. These comparisons help disentangle channel-specific contributions to generation, but making sense of them also requires tracking the marginal prediction error $\mathcal{M}_t$ and the parallelism error $\mathcal{J}_t$ along the trajectories those interventions induce. Since the relevant data conditionals are unavailable for natural text, we construct two proxy measures instead, based on pre-commit model scores computed on the final committed set $\mathcal{S}_t$. These proxies help interpret the controlled comparisons; nevertheless, they do not by themselves demonstrate causal effects, and connecting them to the true errors relies on the assumptions stated below.

Prediction proxy: ***[Devi: proof in appendix]*** Let $\hat{x}_i^{(t)} = \arg\max_v p_\theta(x_i = v \mid X_t)$, $c_i^{(t)} = p_\theta(\hat{x}_i^{(t)} \mid X_t)$, and $H_i^{(t)} = H_{p_\theta}(x_i \mid X_t)$. If the data conditional is a point mass on $\hat{x}_i^{(t)}$, its marginal KL is exactly $-\log c_i^{(t)}$. Moreover,

$$-\log c_i^{(t)} \leq H_i^{(t)}, \tag{8}$$

so $H_i^{(t)} \le \tau$ implies $c_i^{(t)} \ge e^{-\tau}$. This motivates measuring prediction error among lower-entropy commitments:

$$\hat{\mathcal{M}}_t = \sum_{i \in \mathcal{S}_t} (-\log c_i^{(t)}) \mathbf{1}[H_i^{(t)} \le \tau]. \tag{9}$$

Let $A_t^{(\tau)} = \{i \in \mathcal{S}_t : H_i^{(t)} \le \tau\}$. Suppose that, for each $i \in A_t^{(\tau)}$, the data conditional assigns probability at least $1 - \delta_i$ to $\hat{x}_i^{(t)}$, and that $p_\theta(v \mid X_t) \ge \eta_i > 0$ for every data-supported token. Then

$$\mathcal{M}_t(A_t; \theta) \le \hat{\mathcal{M}}_t + \sum_{i \in A_t^{(\tau)}} \delta_i \log \frac{c_i^{(t)}}{\eta_i}. \tag{10}$$

Thus $\hat{\mathcal{M}}_t$ approximates marginal error when the model's low-entropy predictions agree with the data.

Parallelism proxy: The universal dependence bound equation **??** motivates the parallelism proxy $\hat{\mathcal{J}}_t$. For a nonempty selected set, if $|h_i - g_i| \le \varepsilon_i$ at the same history for every $i \in \mathcal{S}_t$, then

$$\mathcal{J}_t(\mathcal{S}_t) \le \hat{\mathcal{J}}_t + \sum_{i \in \mathcal{S}_t} \varepsilon_i - \min_{i \in \mathcal{S}_t} \varepsilon_i. \tag{11}$$

Thus $\hat{\mathcal{J}}_t$ is a useful dependence diagnostic when model and data entropies are close. For each proxy, we sum over decoding steps and average across all evaluation trajectories:

$$\hat{\mathcal{M}} = \mathbb{E}_{\text{eval}}\left[\sum_t \hat{\mathcal{M}}_t^{(\tau)}\right], \qquad \hat{\mathcal{J}} = \mathbb{E}_{\text{eval}}\left[\sum_t \hat{\mathcal{J}}_t\right]. \tag{12}$$

Each scheduler's selection set satisfies the same data-driven bound equation **??**, but the mechanisms vary in how they react to the observable quantity $B_S(g)$. *EB* selects the largest leading segment of the ranking that meets $B_S(g) \le \gamma$, which yields $\mathcal{J}_X(S) \le \gamma + \Delta_S$ where $\Delta_S = B_S(h) - B_S(g)$. With the ranking held fixed, increasing the budget enlarges the selected set; altering the ranking can instead swap which items occupy the selected slots. *PD*, *BD*, *top-k-c*, and *top-k-m* rely on predetermined reveal counts and on confidence or margin driven orderings, either globally or within a block. Adjusting the count causes expansion or contraction when the ranking is fixed, whereas changing the ranking leads to replacements in the selection set. *FDM* screens candidates by confidence and restricts lookahead to a top-K search pool; however, these constraints do not control dependence among elements once committed. *DUS* selects positions by fixed geometric spacing. Spacing alone does not bound $\mathcal{J}_X$, since distant tokens can remain conditionally dependent. A tighter guarantee requires assuming conditional dependence decreases with distance.

4 Experiments and Results

4.1 Setup

Models, Schedulers, datasets, step budgets.

4.2 Validation of Generation Error Proxies

4.3 How Parallelism and Step Budget Affect the Scheduler Performance

4.4 Model-Scheduler Interaction

4.5 Causal Interoperability of Linguistic Structure

5 Related Work

Unmasking schedulers. Early masked DLMs (Sahoo et al., 2024; Nie et al., 2026) revealed the highest-confidence tokens greedily. Block Diffusion (Arriola et al., 2025) imposes semi-autoregressive locality; the EB-Sampler (Ben-Hamu et al., 2026) bounds cumulative entropy to set

how many tokens to reveal; dilated scheduling (Luxembourg et al., 2026) fixes positions geometrically before generation; and FDM (Mo et al., 2025) spends a lookahead pass per step. Theoretical treatments of ordering include Kim et al. (2025) and Zheng et al. (2025), with uncertainty-driven and search-based decoding studied by Chen et al. (2025); Cao et al. (2026) and Luo et al. (2026). A parallel line learns the policy rather than fixing it (Liu et al., 2025; Huang et al., 2025; Jazbec et al., 2026; Chen et al., 2026; Qian et al., 2026).

Limits of parallel generation, and our position. Zhong et al. (2026) characterise the parallelism–quality trade-off; Jiang et al. (2025) show DLMs are provably optimal parallel samplers under conditional independence; and Bansal & Sanghavi (2026) relax that assumption to enable approximate joint sampling. These results bound what *any* scheduler achieves on a *given* model, and the independence they invoke is precisely the quantity our second decision measures. Our question is prior to theirs: how the achievable set moves when the model changes.

6 Conclusion

References

Marianne Arriola, Subham Sekhar Sahoo, Aaron Gokaslan, Zhihan Yang, Zhixuan Qi, Jiaqi Han, Justin T Chiu, and Volodymyr Kuleshov. Block diffusion: Interpolating between autoregressive and diffusion language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=tyEyYT267x.

Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan (eds.), *Advances in Neural Information Processing Systems*, volume 34, pp. 17981–17993. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/958c530554f78bcd8e97125b70e6973d-Paper.pdf.

Parikshit Bansal and Sujay Sanghavi. Enabling approximate joint sampling in diffusion lms, 2026. URL https://arxiv.org/abs/2509.22738.

Heli Ben-Hamu, Itai Gat, Daniel Severo, Niklas Nolte, and Brian Karrer. Accelerated sampling from masked diffusion models via entropy bounded unmasking. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2026. URL https://openreview.net/forum?id=WBcBhT1NKO.

Mingyu Cao, Alvaro H. C. Correia, Christos Louizos, Shiwei Liu, and Lu Yin. Search or accelerate: Confidence-switched position beam search for diffusion language models, 2026. URL https://arxiv.org/abs/2602.10953.

Shirui Chen, Jiantao Jiao, Lillian J. Ratliff, and Banghua Zhu. dultra: Ultra-fast diffusion language models via reinforcement learning, 2026. URL https://arxiv.org/abs/2512.21446.

Ziyu Chen, Xinbei Jiang, Peng Sun, and Tao Lin. Optimizing decoding paths in masked diffusion models by quantifying uncertainty, 2025. URL https://arxiv.org/abs/2512.21336.

Zemin Huang, Yuhang Wang, Zhiyang Chen, and Guo-Jun Qi. Don't settle too early: Self-reflective remasking for diffusion language models, 2025. URL https://arxiv.org/abs/2509.23653.

Metod Jazbec, Theo X. Olausson, Louis Béthune, Pierre Ablin, Michael Kirchhof, João Monteiro, Victor Turrisi, Jason Ramapuram, and Marco Cuturi. Learning unmasking policies for diffusion language models, 2026. URL https://arxiv.org/abs/2512.09106.

Haozhe Jiang, Nika Haghtalab, and Lijie Chen. Diffusion language models are provably optimal parallel samplers, 2025. URL https://arxiv.org/abs/2512.25014.

Jaeyeon Kim, Kulin Shah, Vasilis Kontonis, Sham Kakade, and Sitan Chen. Train for the worst, plan for the best: Understanding token ordering in masked diffusions, 2025. URL https://arxiv.org/abs/2502.06768.

Sulin Liu, Juno Nam, Andrew Campbell, Hannes Stark, Yilun Xu, Tommi Jaakkola, and Rafael Gomez-Bombarelli. Think while you generate: Discrete diffusion with planned denoising. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=MJNywBdSDy.

Ziwei Luo, Ziqi Jin, Lei Wang, Lidong Bing, and Thomas B. Schön. Self-rewarding sequential monte carlo for masked diffusion language models, 2026. URL https://arxiv.org/abs/2602.01849.

Omer Luxembourg, Haim Permuter, and Eliya Nachmani. Plan for speed: Dilated scheduling for masked diffusion language models, 2026. URL https://arxiv.org/abs/2506.19037.

Yichuan Mo, Quan Chen, Mingjie Li, Zeming Wei, and Yisen Wang. Decoding large language diffusion models with foreseeing movement, 2025. URL https://arxiv.org/abs/2512.04135.

Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Large language diffusion models. *Advances in Neural Information Processing Systems*, 38:50608–50646, 2026.

Yu-Yang Qian, Junda Su, Lanxiang Hu, Peiyuan Zhang, Zhijie Deng, Peng Zhao, and Hao Zhang. d3llm: Ultra-fast diffusion llm using pseudo-trajectory distillation, 2026. URL https://arxiv.org/abs/2601.07568.

Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language models. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 130136–130184. Curran Associates, Inc., 2024. doi: 10.52202/079017-4135. URL https://proceedings.neurips.cc/paper_files/paper/2024/file/eb0b13cc515724ab8015bc978fdde0ad-Paper-Conference.pdf.

Jiacheng Ye, Zhihui Xie, Lin Zheng, Jiahui Gao, Zirui Wu, Xin Jiang, Zhenguo Li, and Lingpeng Kong. Dream 7b: Diffusion large language models, 2025. URL https://arxiv.org/abs/2508.15487.

Kaiwen Zheng, Yongxin Chen, Hanzi Mao, Ming-Yu Liu, Jun Zhu, and Qinsheng Zhang. Masked diffusion models are secretly time-agnostic masked models and exploit inaccurate categorical sampling. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://openreview.net/forum?id=CTC7CmirNr.

Yangyang Zhong, Yanmei Gu, Zhengqing Zang, Xiaomeng Li, Yuqi Ding, Xibei Jia, Yuting Shen, Zhenzhong Lan, Liwang Zhu, Weiping Liu, Junlin Zhou, Haisheng Liu, Zhong Xin Yu, Pengxin Luo, Donglian Qi, Yunfeng Yan, and Junbo Zhao. Parallelism and generation order in masked diffusion language models: Limits today, potential tomorrow, 2026. URL https://arxiv.org/abs/2601.15593.

Fengqi Zhu, Rongzhen Wang, Shen Nie, Xiaolu Zhang, Chunwei Wu, Jun Hu, Jun Zhou, Jianfei Chen, Yankai Lin, Ji-Rong Wen, and Chongxuan Li. Llada 1.5: Variance-reduced preference optimization for large language diffusion models, 2025. URL https://arxiv.org/abs/2505.19223.

TABLE OF CONTENTS

CONTENTS

A Training objectives of MDLMs **9**

A.1 LLaDA: masked-diffusion pretraining from scratch 9

A.2 LLaDA-1.5: variance-reduced preference post-training 9

A.3 Dream-7B: autoregressive initialisation and CART 9

B Proof: Decomposition of Generation Error **10**

A TRAINING OBJECTIVES OF MDLMS

We use $x_0 = (x_0^1, \ldots, x_0^L)$ for a clean sequence, x_t for its corrupted version, and $\mathbf{1}[\cdot]$ for an indicator. These objectives determine which conditional prediction problems the denoiser sees during training. They do not specify an inference-time reveal policy or identify the ordering that minimises generation error.

A.1 LLADA: MASKED-DIFFUSION PRETRAINING FROM SCRATCH

LLaDA samples $t \sim \mathcal{U}[0,1]$ and masks every token independently with probability t (Nie et al., 2026). Its mask predictor is trained on masked positions using

$$\mathcal{L}_{\mathrm{LLaDA}}(\theta) = -\mathbb{E}_{\substack{x_0 \sim q,\, t \sim \mathcal{U}[0,1] \\ x_t \sim q_t(\cdot \mid x_0)}} \left[\frac{1}{t} \sum_{i=1}^{L} \mathbf{1}[x_t^i = \mathbf{m}] \log p_\theta(x_0^i \mid x_t) \right]. \tag{13}$$

This is the linear-mask special case of equation **??**: setting the retention probability to $\alpha_t = 1 - t$ gives $-\alpha_t'/(1-\alpha_t) = 1/t$. The corresponding diffusion model satisfies the variational bound

$$-\mathbb{E}_{x_0 \sim q} \log p_\theta(x_0) \le \mathcal{L}_{\mathrm{LLaDA}}(\theta). \tag{14}$$

The 8B-parameter model employs bidirectional attention and was pretrained from the ground up on 2.3 trillion tokens. During supervised fine-tuning, the prompt remains unmasked while the same response-only masked loss is applied to 4.5 million prompt–response pairs. As a result, LLaDA encounters many random masking patterns, but nothing in equation 13 specifies which positions should be revealed jointly at inference time.

A.2 LLADA-1.5: VARIANCE-REDUCED PREFERENCE POST-TRAINING

LLaDA-1.5 is initialised from LLaDA-8B Instruct and then preference-tuned; it is not a separately pretrained diffusion model (Zhu et al., 2025). For prompt x, preferred response y_w, rejected response y_l, current policy π_θ, and frozen reference π_{ref}, ordinary DPO would require exact sequence log likelihoods. Because these are intractable for a MDLM, VRPO uses the diffusion ELBO

$$\mathcal{B}_\pi(y \mid x) = \mathbb{E}_{t \sim \mathcal{U}[0,1]} \mathbb{E}_{y_t \sim q_t(\cdot \mid y,x)} [\ell_\pi(y_t, t, y \mid x)] \le \log \pi(y \mid x) \tag{15}$$

inside the preference score

$$\begin{aligned} s_\theta(x, y_w, y_l) = \beta[&\mathcal{B}_{\pi_\theta}(y_w \mid x) - \mathcal{B}_{\pi_{\mathrm{ref}}}(y_w \mid x)] \\ &- \beta[\mathcal{B}_{\pi_\theta}(y_l \mid x) - \mathcal{B}_{\pi_{\mathrm{ref}}}(y_l \mid x)], \qquad \ell_{\mathrm{VRPO}} = -\log \sigma(s_\theta). \end{aligned} \tag{16}$$

In practice each ELBO is estimated by Monte Carlo. Under independent estimates for y_w and y_l,

$$\mathrm{Var}(\hat{s}_\theta) = \beta^2 \sum_{y \in \{y_w, y_l\}} \left[\mathrm{Var}(\widehat{\mathcal{B}}_{\pi_\theta}(y)) + \mathrm{Var}(\widehat{\mathcal{B}}_{\pi_{\mathrm{ref}}}(y)) - 2\,\mathrm{Cov}\Big(\widehat{\mathcal{B}}_{\pi_\theta}(y), \widehat{\mathcal{B}}_{\pi_{\mathrm{ref}}}(y)\Big) \right]. \tag{17}$$

VRPO lowers this variance by increasing the number of masked samples: it allocates a budget of n samples across n different timesteps, applying one corruption at each timestep, and reuses the same timestep–corruption pair for both the current and reference policies. This final step increases the covariance term in equation 17 when the two ELBO estimates are positively correlated. The published recipe sets $n = 8$, $\beta = 0.2$, and includes the masked-diffusion SFT loss with weight 0.05. This modifies the learned scorer via response-level preference comparisons, rather than training an explicit controller to select positions.

A.3 DREAM-7B: AUTOREGRESSIVE INITIALISATION AND CART

Dream-7B initialises from Qwen2.5 weights and continues training as MDLM on 580 billion tokens (Ye et al., 2025). To preserve the predictive alignment of the autoregressive checkpoint, Dream

retains the shift operation: hidden state h_i predicts position $i+1$ while attention is changed from causal to bidirectional. With $\alpha_t = 1 - t$, its baseline diffusion objective has the same $1/t$ weighting as equation 13.

Dream then substitutes the sequence-level weight with a context- and position-dependent weight via context-adaptive noise rescheduling (CART):

$$\mathcal{L}_{\mathrm{CART}}(\theta) = -\mathbb{E}_{\substack{x_0 \sim q,\, t \sim \mathcal{U}[0,1] \\ x_t \sim q_t(\cdot \mid x_0)}} \sum_{n=1}^{L} \mathbf{1}[x_t^n = \mathbf{m}] w(t, x_t, n) \log p_\theta(x_0^n \mid x_t), \tag{18}$$

$$w(t, x_t, n) = \frac{1}{2} \sum_{i=1}^{L} \mathbf{1}[x_t^i \neq \mathbf{m}] \, \mathrm{Geo}(\rho, |n - i| - 1), \tag{19}$$

where $\rho \in (0, 1]$ determines how quickly a visible token's influence diminishes with distance. A smaller ρ distributes that influence more widely, whereas a larger ρ places greater weight on nearby context. Dream's supervised fine-tuning once more corrupts only the response while keeping the prompt unchanged. CART directly modifies how masked-token losses are weighted based on the visible context, and thus can affect conditional calibration and positional sensitivity. Equations equation 18 and equation 19 continue to specify a denoising objective, not a promise about the optimal reveal order.

B Proof: Decomposition of Generation Error

Let $x = (x_1, \ldots, x_L)$ be distributed according to q over a finite vocabulary. At time t, the history $X_t = ((\mathcal{S}_s, x_{\mathcal{S}_s}))_{s<t}$ records precisely which positions have been revealed so far and their corresponding values. Given this history, a scheduler ϕ selects a subset of the masked positions. We denote its conditional distribution by

$$\pi_{\theta,\phi}(\mathcal{S}_t \mid X_t) := \Pr\big[\phi\big(X_t, p_\theta(\cdot \mid X_t)\big) = \mathcal{S}_t\big] \,. \tag{20}$$

If the scheduler is deterministic, the distribution collapses to a point mass. The sampler then generates their values from $\prod_{l \in \mathcal{S}_t} p_\theta(x_l \mid X_t)$.

Step 1: joint distributions over tokens and schedules. Write $z = (\mathcal{S}_1, \ldots, \mathcal{S}_T)$ and $\pi(\mathcal{S}_t \mid X_t)$ for the scheduling kernel with the controller fixed. Define

$$p_{\theta,\pi}(x, z) = \prod_{t=1}^{T} \left[\pi(\mathcal{S}_t \mid X_t) \prod_{l \in \mathcal{S}_t} p_\theta(x_l \mid X_t) \right], \tag{21}$$

$$q_\pi(x, z) = \prod_{t=1}^{T} \left[\pi(\mathcal{S}_t \mid X_t) q(x_{\mathcal{S}_t} \mid X_t) \right].$$

The same kernel appears in both distributions. Along any valid complete path, the product of data conditionals telescopes to $q(x)$. For a fixed x, the policy probabilities sum to one over paths, hence $\sum_z q_\pi(x, z) = q(x)$. Therefore

$$D_{\mathrm{KL}}\left[q(x) \,\|\, p_{\theta,\pi}(x)\right] \leq D_{\mathrm{KL}}\left[q_\pi(x, z) \,\|\, p_{\theta,\pi}(x, z)\right]. \tag{22}$$

For deterministic π, each x determines a unique path $z_\pi(x)$ under both processes, so the conditional distributions over paths coincide and equality holds. In particular,

$$p_{\theta,\pi}(x) = \prod_t \prod_{l \in \mathcal{S}_t} p_\theta(x_l \mid X_t), \qquad q(x) = \prod_t q(x_{\mathcal{S}_t} \mid X_t), \tag{23}$$

where the histories and selected sets follow $z_\pi(x)$. Adaptivity to previously revealed values does not invalidate this factorisation.

Step 2: conditional KL at each history. For deterministic π, substitute equation 23 into the sequence KL and condition on the history at each step:

$$D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\pi}\right] = \sum_t \mathbb{E}_{q,\pi}\left[D_{\mathrm{KL}}\left[q(x_{\mathcal{S}_t} \mid X_t) \,\|\, \textstyle\prod_{l \in \mathcal{S}_t} p_\theta(x_l \mid X_t)\right]\right]. \tag{24}$$

Here $\mathbb{E}_{q,\pi}$ means draw $x \sim q$ and apply the controller to the successively revealed data values. The conditional divergences inside the expectation depend on the realised history; they are not themselves history-averaged quantities.

Step 3: split the block error. Fix $X = X_t$ and $\mathcal{S} = \mathcal{S}_t$, and insert $\tilde{q}(x_{\mathcal{S}} \mid X) = \prod_{l \in \mathcal{S}} q(x_l \mid X)$ into the log ratio:

$$\begin{aligned} D_{\mathrm{KL}}\left[q(x_{\mathcal{S}} \mid X) \,\|\, \textstyle\prod_{l \in \mathcal{S}} p_\theta(x_l \mid X)\right] &= D_{\mathrm{KL}}\left[q(x_{\mathcal{S}} \mid X) \,\|\, \tilde{q}(x_{\mathcal{S}} \mid X)\right] \\ &\quad + \sum_{l \in \mathcal{S}} \mathbb{E}_{q(x_{\mathcal{S}} \mid X)}\left[\log \frac{q(x_l \mid X)}{p_\theta(x_l \mid X)}\right] \\ &= \mathcal{J}_t(\mathcal{S}) + \sum_{l \in \mathcal{S}} D_{\mathrm{KL}}\left[q(x_l \mid X) \,\|\, p_\theta(x_l \mid X)\right] \\ &= \mathcal{J}_t(\mathcal{S}) + \mathcal{M}_t(\mathcal{S}; \theta). \end{aligned} \tag{25}$$

Each log ratio depends only on x_l, so its joint expectation equals its marginal expectation. Substituting into equation 24 proves Proposition **??**. □

Stochastic policies. For a stochastic scheduling kernel, cancellation of the policy factors in the joint KL gives

$$D_{\mathrm{KL}}\left[q \,\|\, p_{\theta,\pi}\right] \le \mathbb{E}_{q_\pi}\left[\sum_t \big(\mathcal{M}_t(\mathcal{S}_t; \theta) + \mathcal{J}_t(\mathcal{S}_t)\big)\right]. \tag{26}$$

By the KL chain rule, the right-hand side minus the left-hand side is $\mathbb{E}_{x \sim q} D_{\mathrm{KL}}\left[q_\pi(z \mid x) \,\|\, p_{\theta,\pi}(z \mid x)\right]$. Thus equality holds exactly when these posterior path distributions agree q-almost surely. Conditioning on an external random seed can recover a deterministic-policy equality; averaging such conditional errors need not equal the KL of the seed-marginalised generator.